

# Breeding Diverse Noise: A Population-Based Companion to “It’s Never Too Late” Collapse Recovery

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Code: [github.com/AniAggarwal/divgen](https://github.com/AniAggarwal/divgen) (fork of the official implementation)

## Abstract

Harrington et al.’s “It’s Never Too Late” makes a lovely observation: mode collapse in trained text-to-image models is not a fact about the model so much as a fact about *where we start it* — and the initial noise can simply be optimized, end-to-end, into a diverse set of generations. This note is written in appreciation of that work, and asks one follow-up question of it: does the noise need a *gradient* to find its way? We reformulate their objective as a two-objective *artificial-selection* search<sup>1</sup> over *sets* of initial noises, using the multi-objective machinery from our ECAD work on caching schedules. On GenEval with SDXL-Turbo, breeding reaches the gradient-based optimization’s numbers on its own metrics (slightly ahead on three of the four reported, slightly behind on LPIPS) — without a single backward pass through the sampler. We describe what the population view adds (memory-free scaling to large models, non-differentiable set-level objectives, an explicit quality–diversity Pareto front, self-tuned step sizes) and what it costs, and offer it as a companion, not a replacement, to the original method.

## 1 Appreciation, and a question

Three ideas in Harrington et al. do a lot of work. First, that inference-time diversity is recoverable at all — that a collapsed model still *contains* its diversity, reachable through the initial noise  $x_0$ . Second, the frequency analysis (their Fig. 9): optimization acts almost entirely on the lowest third of the noise spectrum, which both explains the method and immediately yields the pink-noise initialization that improves every method they test, including the baselines. Third, the observation that *set-level* objectives (DPP, Vendi) are the right currency for diversity, because they cannot be

<sup>1</sup>We deliberately avoid the customary term “evolutionary algorithm.” Natural evolution optimizes no objective — it is selection without a selector, purposeless by construction — so attaching the word to an explicitly objective-driven procedure seems to us a misnomer. What this algorithm actually performs is closer to what breeders have always done: *artificial selection* toward a stated goal. We therefore speak of *breeding* noise, and of selection-based or population-based search.

gamed by making one image strange — a point their user study makes empirically.

The method that operationalizes these ideas is gradient descent through the frozen sampler: differentiable diversity and quality scores are backpropagated to the noise batch. Our question is narrow: *is the gradient essential, or is it one search procedure among several for the landscape their analysis reveals?* The question matters practically: back-propagating through a 10B-parameter flow model is a memory wall on commodity hardware, and the most human-aligned objectives in their study (DPP and Vendi at the set level) are exactly the ones where differentiability is most awkward.

We answer it by transplanting the multi-objective search we built for ECAD [2] — which breeds diffusion *caching schedules* against a quality–speed front — onto their problem: same NSGA-II [3] engine, same two-objective structure, with the chromosome swapped from binary cache decisions to a set of noise tensors.

## 2 Breeding noise sets

**Individuals.** Diversity in the source paper is measured over a set of  $B=4$  images per prompt, so the unit of selection is the whole set: an individual is  $\{x_0^{(1)}, \dots, x_0^{(B)}\}$ , plus two *strategy genes* (a mutation step  $\sigma$  and a per-element mutation rate  $\rho$ ) that are inherited and perturbed alongside the noise, as in classical self-adaptive strategies [4].

**Objectives.** Each individual is rendered (forward passes only) and scored with the paper’s own code: mean per-image CLIPScore as *quality*, and a set-level statistic — patchwise DINOv2 distance, DPP log-det, or Vendi — as *diversity*. NSGA-II maintains the Pareto front of the two, exactly as ECAD maintains quality–versus–speed. Generation 0 is drawn i.i.d. from the prior, so the paper’s baseline is embedded in every run as the starting population.

**Operators, each borrowed from a finding of theirs.** Mutations are spectrally colored by  $1/(1+f)^\beta$ , so proposals move low frequencies more than high ones — their Fig. 9

**Table 1:** GenEval, SDXL-Turbo, white-noise initialization, set size 4. Paper rows from Harrington et al. Tab. 1 (their eval harness); our rows use the official repository’s own metric code over all 553 prompts in its GenEval list (population 40; up to 60 generations, and prompts still below the reference diversity receive one further 90-generation pass). Diversity metrics: higher is better.

| Method                       | DINO         | DreamSim     | LPIPS | CLIP  |
|------------------------------|--------------|--------------|-------|-------|
| i.i.d.                       | 0.588        | 0.249        | 0.642 | 0.335 |
| Parmar et al. [5]            | 0.705        | 0.331        | 0.682 | 0.333 |
| Harrington et al. (gradient) | 0.784        | 0.411        | 0.767 | 0.349 |
| Bred, gen. 0 (i.i.d.)        | 0.658        | 0.298        | 0.670 | 0.374 |
| Bred, final                  | <b>0.790</b> | <b>0.437</b> | 0.745 | 0.393 |

dynamic, made into an operator (and the same filter behind their pink-noise initialization). Their  $\chi_d$ -norm regularizer  $K(\epsilon)$ , which keeps noises in the high-density shell of the prior, becomes a *repair* step: every latent is re-standardized after every operator, enforcing the constraint exactly rather than penalizing its violation. Crossover exchanges whole noise slots between parent sets — recombination in the space of sets, respecting that individual noises are the atoms of diversity.

**Self-tuned exploration.** In our ECAD ablations, a 1% mutation rate stalls in poor local optima and 15% converges too slowly; the sweet spot was never known. Here each individual carries its own  $(\sigma, \rho)$ , and selection tunes them: the population raises  $\sigma$  early (exploration) and anneals it toward 0.05 as it converges (exploitation), with no schedule supplied by us (Fig. 1, right). A controlled comparison (Tab. 2, top) confirms the 1% rate stalls here too, while the aggressive 15% rate buys diversity at a visible quality cost; the self-tuned variant holds the highest quality at comparable diversity, with no rate chosen in advance. The same table shows that removing the low-frequency mutation bias ( $\beta=0$ ) loses diversity with no quality gain, while strengthening it ( $\beta=2$ ) buys large diversity gains at modest quality cost — both consistent with the source paper’s spectral analysis.

### 3 Results

Table 1 gives the headline. Two sanity checks first: our generation-0 population, scored with the official code, reproduces the paper’s i.i.d. row almost exactly, and quality *rises* during search rather than being traded away. Across the reported metrics, breeding lands slightly ahead of the gradient optimizer on DINO, DreamSim, and CLIPScore, and slightly behind on LPIPS — despite never computing  $\partial(\text{image})/\partial x_0$ . We read this as evidence for a claim implicit in the original paper: the low-frequency structure of the noise landscape is benign enough that *any* sensible search finds it; the gradient is one excellent way, not the

**Table 2:** Design ablations (six GenEval prompts, 45 generations, population 32, matched seeds; defaults marked  $\star$ ; gen-0 i.i.d. reference: DINO 0.639, CLIP 0.338). Variants move *along* the quality–diversity front: the defaults hold the highest quality, while removing the spectral bias ( $\beta=0$ ) or fixing a 1% rate is dominated outright.

| Group           | Variant            | DINO         | DreamSim     | CLIP         |
|-----------------|--------------------|--------------|--------------|--------------|
| mutation rate   | fixed 1%           | 0.705        | 0.324        | 0.346        |
|                 | fixed 15%          | 0.790        | 0.464        | 0.336        |
|                 | self-tuned $\star$ | 0.757        | 0.361        | <b>0.347</b> |
| spectral bias   | $\beta=0$ (white)  | 0.714        | 0.316        | 0.348        |
|                 | $\beta=1$ $\star$  | 0.757        | 0.361        | 0.347        |
|                 | $\beta=2$          | <b>0.830</b> | <b>0.593</b> | 0.339        |
| crossover       | off                | 0.802        | 0.529        | 0.333        |
|                 | set-level $\star$  | 0.757        | 0.361        | 0.347        |
| $\chi_d$ repair | off                | 0.750        | 0.438        | 0.342        |
|                 | every op. $\star$  | 0.757        | 0.361        | 0.347        |
| initialization  | white $\star$      | 0.757        | 0.361        | 0.347        |
|                 | pink $\alpha=0.2$  | 0.781        | 0.428        | 0.341        |

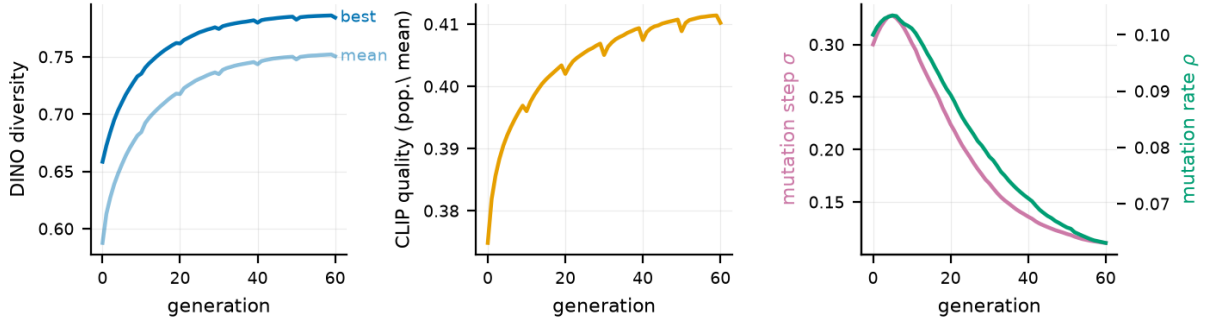
only one.

Qualitatively the recovery is the same phenomenon their Fig. 1 shows: four interchangeable grey park benches become a teal bench, a golden-hour park, new viewpoints — every image still plainly a bench (Fig. 2).

**Two extensions the population view invites.** (i) *Prompt-typed islands.* Sub-populations bred on disjoint prompt distributions (painted landscapes, five-word prompts, long GPT-written prompts), with the best noise sets migrating between islands every few generations. Because a noise set is prompt-agnostic, a migrant that keeps winning after crossing prompt types is one whose diversity *generalizes*: in our runs, migrated elites retain 0.65–0.69 DINO diversity on prompt families they never saw during selection. Per-prompt gradient descent has no natural analogue of this test. (ii) *Searching program space.* Instead of breeding the noise tensor, we can breed a small symbolic *program* (compositions of white noise, pink filtering, and low-frequency gratings) that generates it. Program search reaches the highest raw diversity we observe (DINO 0.886) at a real cost in prompt fidelity — it maps the far end of the quality–diversity front — and its early winners rediscover low-frequency structure on their own, an independent confirmation of the paper’s spectral analysis.

### 4 What each method offers

We want to be precise about the trade, because the comparison flatters neither method alone. The gradient optimizer is far more sample-efficient where it applies: at 0.345 s per iteration on an A100 (their Sec. 4.2), it reaches its result in seconds per prompt, while breeding spends thou-



**Figure 1:** Convergence averaged over GenEval prompts (SDXL-Turbo, population 40). Left: the bred diversity objective. Middle: CLIP quality *rises* during search — the Pareto selection concedes nothing for the diversity gain. Right: the self-adaptive mutation step  $\sigma$  and rate  $\rho$ , raised then annealed by selection alone.



**Figure 2:** “A photo of a bench,” SDXL-Turbo. Left: i.i.d. generation 0. Right: the bred set.

sands of forward renders to arrive at the same neighborhood. Population search earns that cost back in three situations. *Memory*: it never backpropagates through the sampler, so model scale adds no optimizer memory — relevant exactly at the FLUX scale the paper targets. *Objectives*: non-differentiable scores — true DPP samplers, human ratings, detector-based prompt-faithfulness checks — enter the loss with no relaxation, and it is precisely such set-level scores that their user study found most human-aligned. *Output*: a run yields an explicit Pareto *front* of noise sets, so the quality–diversity trade-off becomes a menu rather than a weight to tune. The two approaches also compose naturally: bred noise is a warm start for gradient refinement, and their pink-noise initialization already improves our generation 0 the same way it improves theirs.

## 5 Closing

Each of the original paper’s findings translated directly into a design decision here: the frequency analysis told us what mutations to propose, the norm regularizer told us what constraint to repair to, and the set-level metrics told us what to select for. We offer the population view back as a companion — same landscape, second vehicle — with code, con-

figurations, and scaling scripts in the fork above, in the hope it is useful to the authors’ future work on collapse recovery.

## References

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